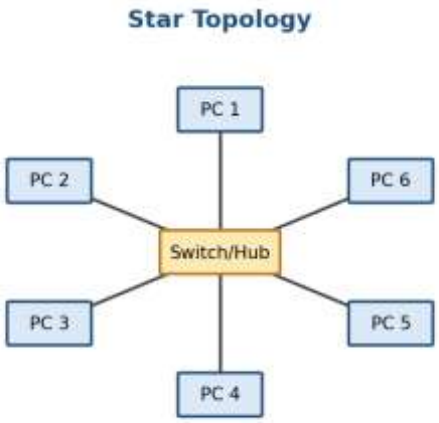
Q1. Star, Bus and Mesh Topology Diagrams

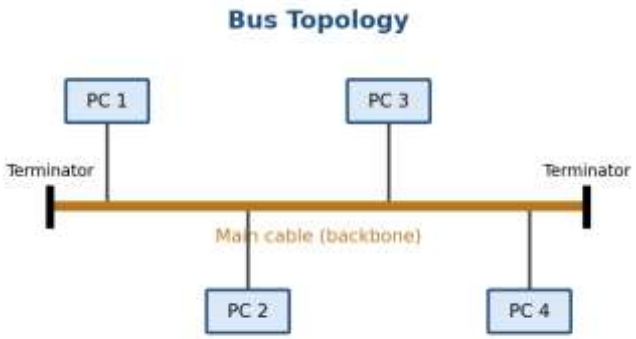


Star Topology

Labels: PC 1–PC 6 (nodes) connected to a central Switch/Hub.

App example: Zomato.

Every customer, restaurant and delivery partner connects to Zomato's central servers, so orders and payments are managed in one place.

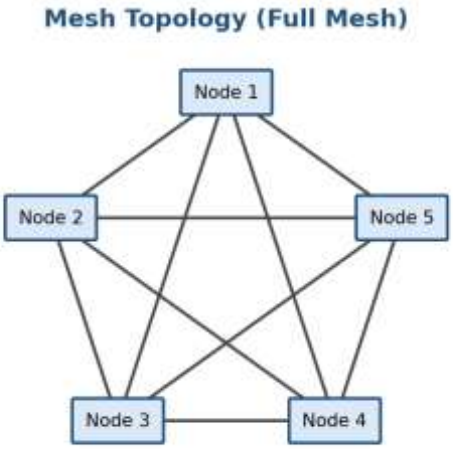


Bus Topology

Labels: PC 1–PC 4 (nodes) attached to a single main cable (backbone), with a terminator at each end.

App example: IPL live score ticker (broadcast style).

One source sends the same data along a shared line and every connected device just listens. (Bus is rare today and mostly used in very small or old networks.)



Mesh Topology

Labels: Node 1–Node 5, each connected directly to every other node.

App example: WhatsApp.

WhatsApp's data centres are connected to each other with many links, so if one path fails your message automatically takes another and is still delivered.

Q2. Comparison of Star, Bus, Ring and Mesh

Topology	Pros	Cons	Best Use Case (with app example)
Star	• Easy to install and manage • One cable fault affects only one device • Easy to add or remove devices • Good performance, no data collisions	• Central switch is a single point of failure • Needs more cable than bus • Hub/switch adds cost • Network speed depends on the central device	Offices, labs, homes (Wi-Fi router). Example: Zomato – every user's app connects to Zomato's central servers.
Bus	• Cheap, needs the least cable • Simple to set up • Good for very small networks • Easy to extend by joining cable	• Main cable failure stops the whole network • Slows down as devices increase • Hard to find faults • Data collisions are common	Very small, temporary LANs. Example: rarely used today; Instagram, Zomato etc. do not use it.
Ring	• Orderly data flow, few collisions • Equal access for every device • Works well under heavy load • No central device needed	• One broken device or cable can stop the ring • Hard to troubleshoot • Adding a device disturbs the network • Data passes through every node, so it can be slow	Telecom and ISP fibre backbones in cities. Example: Paytm/UPI traffic travelling over an ISP's fibre ring.

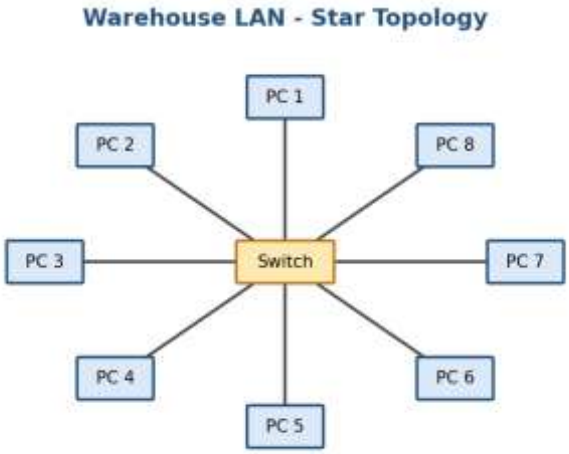
Topology	Pros	Cons	Best Use Case (with app example)
Mesh	• Very reliable, many alternate paths • One link failure does not stop traffic • Fast, dedicated links • Easy to find faulty links	• Very expensive (too many cables/ports) • Complex to install and manage • Needs many connections per device (n(n-1)/2 links) • Hard to expand	Critical networks that must never go down. Example: Instagram/WhatsApp data centres linked to each other so messages reroute if one fails.

Q3. Topology for a Multiplayer Game (PUBG / Free Fire)

Chosen topology: Star (client–server) between players and the game server, with the game servers of different regions linked in a mesh.

Every player's phone connects to a nearby central game server, which controls the match, so all players see the same game state and cheating is hard. This keeps speed (low ping) good because each player has only one connection to manage. Reliability is high because the linked regional servers can take over if one server fails. Cost is moderate: far lower than a full mesh between all players, and servers can be added only where players are many.

Q4. Warehouse Network for 8 Computers



Recommended topology: Star topology using one network switch (a 16-port switch leaves room to grow).

It is low-cost and easy to set up for 8 computers. If one cable or PC fails, the others keep working, so troubleshooting is quick. New computers or a printer can simply be plugged into a free switch port as the warehouse grows.